

Reducing the “Drag” of GPU Memory Accesses with Slipstreaming

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Summary

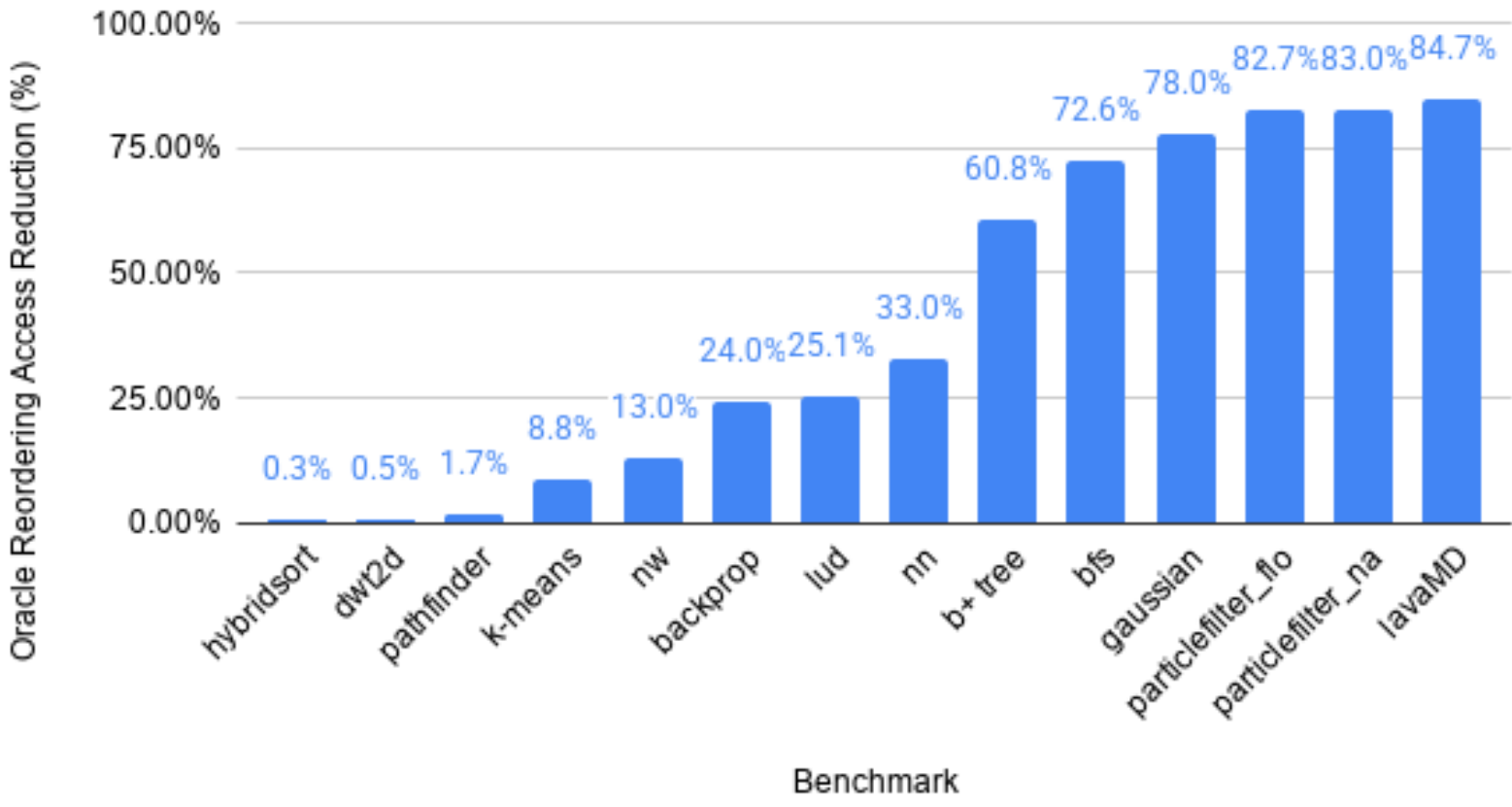
Memory bottlenecks limit GPU throughput, even with conventional warp-level coalescing and cache hierarchies. Slipstreaming targets an additional source of locality: cache lines shared across contiguous warps, or clusters of warps, in the same SM.

Slipstreaming combines a clustering-aware warp scheduler with an L1 cache replacement policy that protects lines likely to be reused by nearby warps. This emulates a form of inter-warp coalescing, reducing expensive L2 and global memory accesses.

Across evaluated workloads, Slipstreaming improves GPU-wide IPC by 2.72% geomean while requiring only 160 KB of additional state (<1% L2 on QV100).

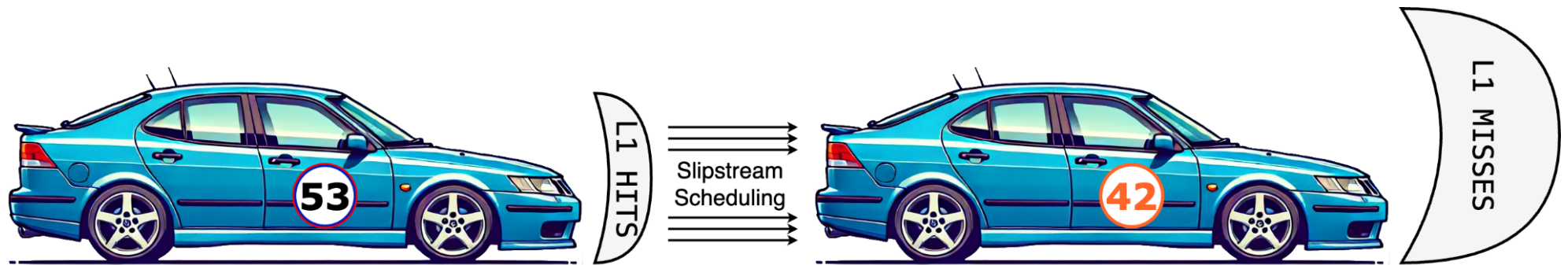
Motivation

Oracle-Reordered Memory Access Reduction Potential



Oracle reordering reveals redundant cache-line accesses across threads, indicating exploitable inter-warp locality beyond conventional warp-level coalescing.

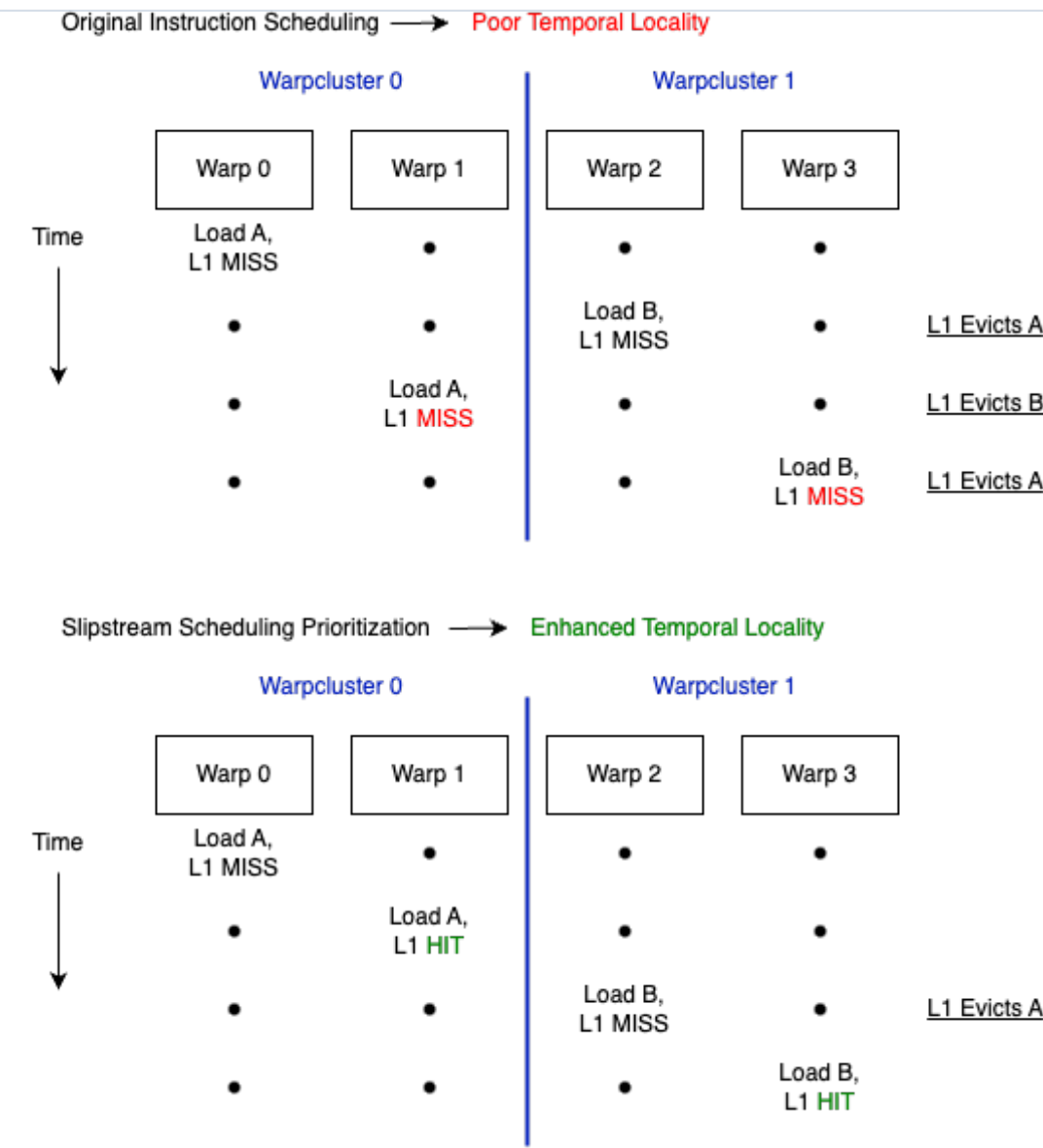
Slipstream Intuition



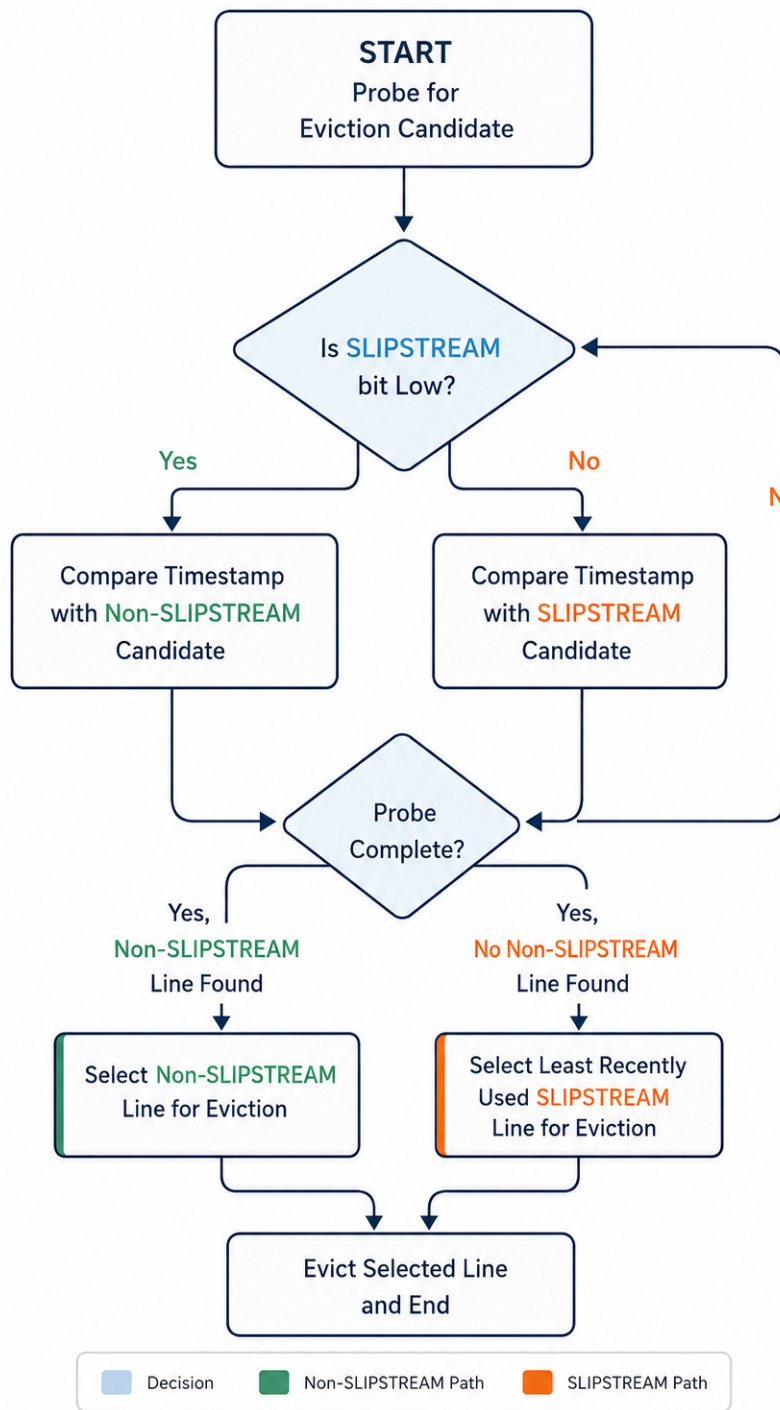
The leader pays the L1 miss; nearby follower warps can hit in L1 instead of missing again.

Scheduling Strategically

Scheduling loads strategically can reduce cache misses without increasing cache size.



SC-LRU Replacement Policy



SC-LRU preferentially evicts non-Slipstream lines and only evicts a Slipstream line if no non-Slipstream candidate is available.

SC-LRR Prioritization

SC-LRR Scheduling Priority Tiers

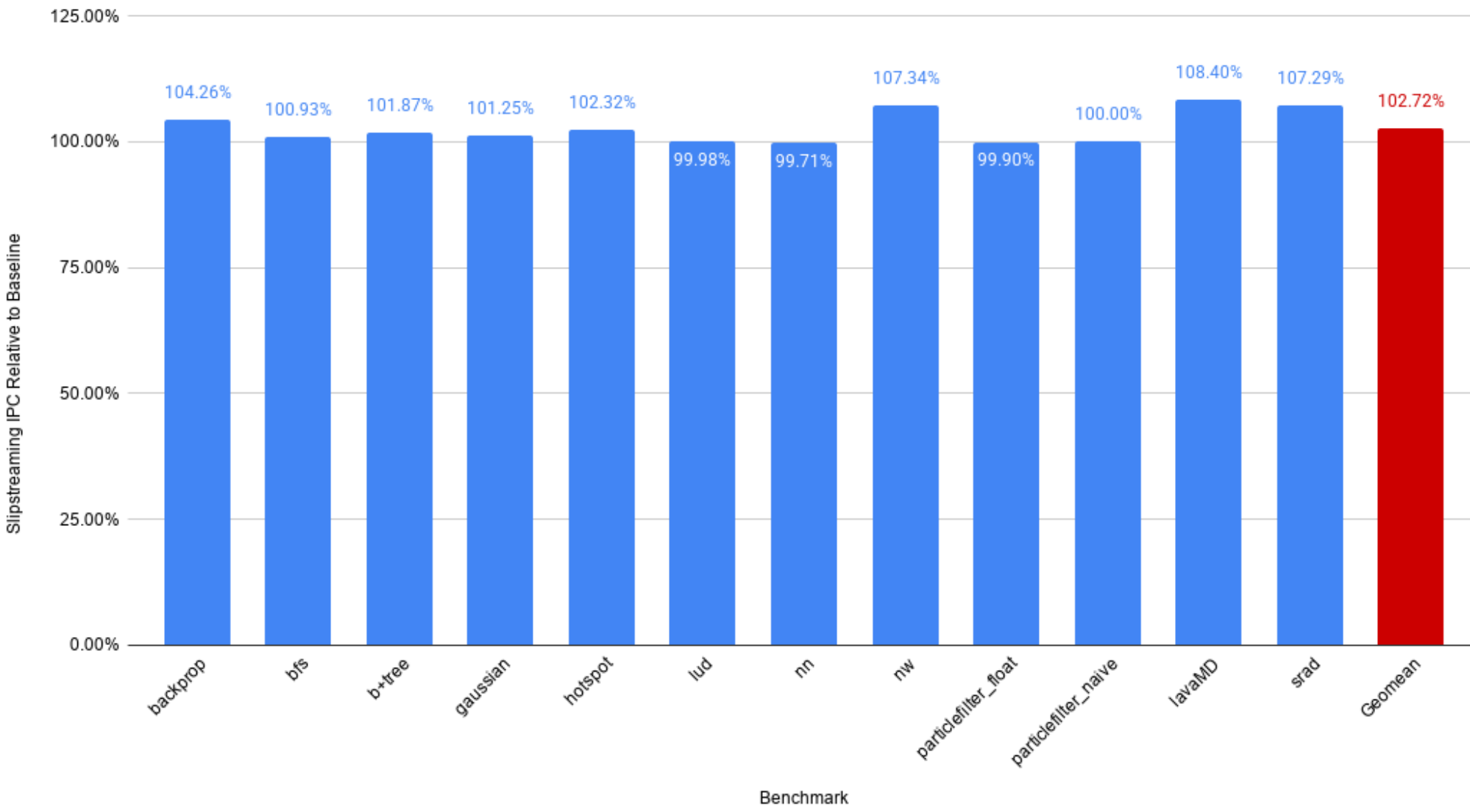
- 1 PRIMARY-SLIPSTREAM-NOT-REACHED**
Warp is in the primary warpcluster and has not yet reached the primary LD/ST instance. Highest priority.
- 2 SLIPSTREAM-TRACKED-NOT-REACHED**
Warp is tracked by the slipstream engine, not in the primary warpcluster, and has not yet reached the tracked LD/ST instance.
- 3 NON-SLIPSTREAM**
Warp is not currently tracked by the slipstream engine.
- 4 SLIPSTREAM-TRACKED-REACHED**
Warp is tracked by the slipstream engine, not in the primary warpcluster, and has already reached the tracked LD/ST instance.
- 5 PRIMARY-SLIPSTREAM-REACHED**
Warp is in the primary warpcluster and has already reached the primary LD/ST instance. Lowest priority.

Methodology

Implemented in Accel-sim (GPGPU-sim 4.0) and evaluated via SASS trace-based simulation of executing the Rodinia 3.1 benchmark suite on an NVIDIA QV100

Results

Slipstreaming Scheduling and Replacement Policy IPC Relative to Loose Round Robin and LRU



Geomean IPC improvement: +2.72%

Peak speedup: +8.40%

Hardware Cost

Slipstream Entry – 32 / Warpcluster

Slipstream-ID	Slipstream Gap	Bitmask	PC	Total
5b	10b	8b	32b	7B

Warpcluster Subtable – 2 / SM

Slipstream Entries	Cluster ID	Primary Slipstream ID	Total
224B	9b	6b	226B

Line Map Entry – 1024 / SM

Slipstream Cluster 0 Entry	Slipstream Cluster 1 Entry	Total
5b	5b	1 . 25B

Slipstreaming adds approximately 160 KB of GPU-wide state, a modest hardware overhead.

Takeaways

- Targets latent inter-warp locality beyond conventional coalescing.
- Co-designs scheduling and L1 replacement to preserve useful cache lines.
- Improves throughput with modest hardware cost.

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